Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam B · Form 1: Single-step application

Multiple Choice Questions: Select the best answer. (1 point each)

1. A 3 N horizontal push does not move a crate; maximum static friction is 8 N. What is actual friction magnitude?

(A) 11 N

(B) 3 N

(C) 8 N

(D) 5 N

2. A 40 kg student stands on a scale in an elevator that is moving up and accelerating up, with acceleration magnitude 1.5 m/s². What force does the scale read? Use g = 9.8 m/s².

(A) 226 N

(B) 452 N

(C) -452 N

(D) 904 N

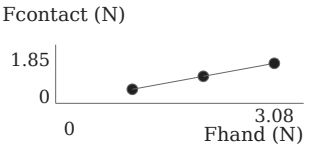
3. Touching 2 kg and 3 kg carts accelerate on a frictionless floor when a hand pushes the 2 kg cart right. Fhand (N) -> Fcontact (N): (1.08, 0.648); (2.08, 1.248); (3.08, 1.848). What signed vertical change follows a decrease of 0.5 horizontally?

(A) -0.3

(B) -0.15

(C) 0

(D) -0.45



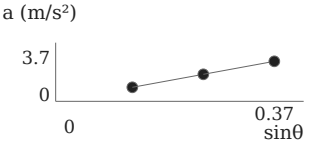
4. Release a block on a frictionless ramp and record downhill acceleration. Use g=10 m/s². sinθ -> a (m/s²): (0.1296, 1.296); (0.2496, 2.496); (0.3696, 3.696). Predict the reading at horizontal coordinate 0.3096.

(A) 1.858

(B) 5.573

(C) 4.334

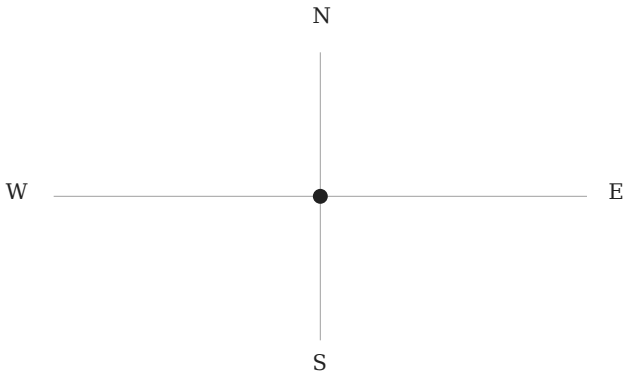
(D) 3.096



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Two students push a cart of mass 10 kg on a frictionless horizontal floor. A exerts 220 N at 15° north of east. B exerts 310 N at 33° south of west.

a. Draw and label the actual forces acting in the plane of the floor on the dot. Draw whole vectors, not components. **(1 pt)**



b. Show net-force x and y components, then find acceleration magnitude and direction using compass words. **(3 pt)**

6. A 50 kg passenger stands on a scale in an elevator. Up is positive; use g=10 m/s². a (m/s²) -> N (N): (1.08, 554); (2.08, 604); (3.08, 654).

a. Build the model for passenger on a scale from physics and predict both slope and intercept. **(2 pt)**

b. Label the axes and units and demonstrate interpolation between two readings. **(2 pt)**

7. A mass-m cart is pushed right by force F at θ below horizontal. It slides right with kinetic friction coefficient μ. Derive horizontal acceleration with zero vertical acceleration. **(3 pt)**